

Original article

Development of colchicine-infused medicated gel with natural permeation enhancers for rheumatoid arthritis treatment

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Author contributions

Dr. Yogesh V Ushir conceived the presented idea. Kiran A. Suryavanshi developed the theory and performed the experiments, calculations and analysed the data, wrote the manuscript. Kundan J. Tiwari contributed to the final version of the manuscript. All authors edited the manuscript, figures, and graphs and contributed in approval of the final version of the article.

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Conflicts of interest

Authors declare no conflicts of interest.

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Development of colchicine-infused medicated gel with natural permeation enhancers for rheumatoid arthritis treatment

Abstract

Rheumatoid arthritis (RA) is a chronic inflammatory condition requiring effective therapeutic approaches to improve patient outcomes and quality of life. Topical anti-inflammatory treatments offer a promising alternative to conventional oral therapies by delivering targeted relief at the inflammation site while reducing systemic side effects. This study aimed to develop a colchicine-loaded gel incorporating natural permeation enhancers to enhance transdermal drug delivery for RA management. Using a Quality by Design (QbD) approach, the formulation was systematically optimized for ideal viscosity and drug release profiles. Colchicine was formulated into a gel with natural permeation enhancers, particularly *Simmondsia chinensis* oil, and subjected to comprehensive evaluations. These included *in vitro* drug release, *ex vivo* skin diffusion, and *in vivo* pharmacokinetics in Wistar rats. The optimized formulation, SL06, contained 1.2 g of carbopol 940 and 0.9 mL of *Simmondsia chinensis* oil, demonstrating superior drug permeation and sustained release compared to standard colchicine. Pharmacokinetic analysis revealed enhanced bioavailability due to improved skin penetration facilitated by the natural permeation enhancer. The findings highlight the potential of a colchicine-loaded gel for RA treatment, offering enhanced drug delivery and superior plasma concentrations. This novel formulation holds promise for safer and more effective RA management.

KEYWORDS: Rheumatoid arthritis, Colchicine, *Simmondsia chinensis*, Gel, Permeation enhancers, *ex-vivo*, *in-vivo*.

Running title: Colchicine-Infused Gel with Natural Enhancers for RA Treatment

1. Introduction

Rheumatoid arthritis is a chronic, systemic autoimmune disease characterized by inflammation that primarily targets the joints and surrounding soft tissues [1]. Rheumatoid arthritis affects approximately 1% of the global population and is more prevalent in females than in males [2]. The pathology of RA involves the immune system mistakenly attacking the body's own tissues, leading to damage of the synovium, cartilage, and bone within the joints [3]. Current treatment approaches for RA primarily include nonsteroidal anti-inflammatory drugs (NSAIDs), disease-

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modifying antirheumatic drugs (DMARDs), and biologic agents [4,5]. While these medications can help manage symptoms and slow disease progression, they are often associated with notable drawbacks, including gastrointestinal side effects, immunosuppression, and the necessity for regular monitoring [6,7]. Consequently, developing alternative drug delivery systems that provide localized and sustained therapeutic effects is crucial for enhancing patient outcomes and minimizing systemic side effects.

Transdermal drug administration presents a promising alternative to traditional treatments, offering targeted, site-specific action and improved patient compliance. This approach enhances skin permeability and enables sustained drug release, both of which contribute to better therapeutic outcomes and greater adherence to treatment [8,9]. Topical gels are favoured as drug delivery systems due to their non-greasy texture, ease of removal, and prolonged retention at the application site. Structurally, they consist of a 3D matrix where colloidal particles are uniformly dispersed in a solvent, allowing for effective and sustained delivery [10]. However, the skin's stratum corneum serves as a natural barrier, limiting drug permeability and necessitating the incorporation of permeation enhancers to facilitate drug absorption.

Colchicine, a potent anti-inflammatory agent, is effective in reducing joint inflammation and regulating immune responses in rheumatoid arthritis RA. It works through multiple mechanisms, including the disruption of microtubule formation, which inhibits leukocyte movement and inflammatory cell functions. Additionally, colchicine reduces the production of pro-inflammatory cytokines, suppresses the NLRP3 inflammasome, and limits neutrophil activation and migration. It also decreases the production of reactive oxygen species (ROS), reducing oxidative stress and preventing tissue damage. Together, these actions help control inflammation, manage disease activity, and prevent joint damage in RA [11,12].

The Quality by Design (QbD) model promotes a systematic approach to pharmaceutical product development, beginning with predefined specifications based on product and process knowledge as well as process control. This framework ensures the quality of the final product by focusing on thorough understanding and control from the outset [13]. In this study, formulation variables were examined using a QbD-based approach to develop a colchicine gel, utilizing Carbopol 940 as the gel-forming agent and *Simmondsia chinensis* oil as a natural permeation enhancer. Employing natural permeation enhancers supports patient safety by reducing reliance on synthetic alternatives and aligns with the rising interest in biocompatible and eco-friendly pharmaceutical excipients. The formulated gels were evaluated through *in vitro* drug release, *ex vivo* drug diffusion, and *in vivo* studies.

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usefulness of natural enhancer *Simmondsia chinensis*

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section: Identification of critical quality attributes

2. Materials and methods

2.1. Materials

Colchicine was generously provided as a gift sample by Vital Laboratory PVT LTD, Vapi, Gujrat, India. *Simmondsia chinensis* (Jojoba seeds) were procured from a local market in Jaipur, Rajasthan, India. Carbopol-940 was sourced from Leonid Chemicals Pvt Ltd, Bangalore. Additional materials and reagents used in this study included Polyethylene glycol, Isopropyl alcohol, Methyl Paraben, Propyl Paraben, and Triethanolamine, all obtained from Loba Chemicals, Mumbai. All chemicals and reagents used in this research were of analytical grade.

2.2. Extraction of oil from *Simmondsia chinensis*

Simmondsia chinensis nuts (2.5 Kg) were extracted with methanol three times over a 24-hour period at a controlled room temperature of 20–25°C. The methanol solvent was then evaporated at 50°C under reduced pressure, yielding a concentrated methanol extract (280 g). This extract was fractionated with n-hexane and water, resulting in an n-hexane fraction (86 g) and a residual aqueous layer. The aqueous layer was subsequently extracted with ethyl acetate, producing an ethyl acetate fraction (120 g) and a water fraction (90 g). The ethyl acetate fraction, which exhibited the highest bioactivity, was further purified through chromatography on a Wakogel C200 column (Wako Pure Chemical, Japan) using a gradient mixture of n-hexane, ethyl acetate, and methanol under an applied voltage. The primary bioactive components were isolated and further purified with a sulfuric acid-ethanol solution in ratio of 1:9.

2.3. Identification of critical material attributes

The inactive components of the gel formulation consisted of an oil forming the oil phase and a thickening agent. Key excipients were selected as critical material factors, specifically the concentration of the oil (*Simmondsia chinensis* oil) and the concentration of the thickening agent (Carbopol 940) [14].

2.4. Identification of critical quality attributes

A colchicine-loaded gel for treating rheumatoid arthritis was developed and optimized using Quality by Design (QbD) methodologies. Literature reviews identified viscosity and

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cumulative drug release percentage as critical quality attributes essential to the gel's performance [15].

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2.5. Preparation of gel

Carbopol 940 was gradually dissolved in 50% distilled water with continuous stirring for 1 hour to prevent agglomeration. Separately, triethanolamine was dissolved in 10 mL of distilled water and stirred for 10 minutes. The triethanolamine solution was then added to the Carbopol solution, and the pH was adjusted to 7.4 by stirring for an additional 10 minutes. In a separate beaker, isopropyl alcohol (IPA), polyethylene glycol (PEG), and the required quantity of extracted oil were combined. This mixture was stirred on a mechanical shaker at 50 rpm until the oil was fully miscible with the aid of IPA and PEG. Finally, this mixture was gradually added to the Carbopol 940 solution, ensuring thorough blending of both phases [16]. Detailed formulation compositions are provided in Table 1.

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Table 1. Composition of colchicine-loaded gel with natural permeation enhancers

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Gel Code	Drug (%)	Carbopol 940 (gm)	<i>Simmondsia chinensis</i> Oil (ml)	PEG (ml)	IPA (ml)	Methyl Paraben (gm)	Propyl Paraben (gm)	TEA (ml)	DW (100 g)
SC1	1	0.5	0.1	5	5	0.02	0.01	q. s.	q. s.
SC2	1	0.5	0.5	5	5	0.02	0.01	q. s.	q. s.
SC3	1	0.5	0.9	5	5	0.02	0.01	q. s.	q. s.
SC4	1	1.2	0.1	5	5	0.02	0.01	q. s.	q. s.
SC5	1	1.2	0.5	5	5	0.02	0.01	q. s.	q. s.
SC6	1	1.2	0.9	5	5	0.02	0.01	q. s.	q. s.
SC7	1	1.9	0.1	5	5	0.02	0.01	q. s.	q. s.
SC8	1	1.9	0.5	5	5	0.02	0.01	q. s.	q. s.
SC9	1	1.9	0.9	5	5	0.02	0.01	q. s.	q. s.

2.6. Evaluation of Gel

2.6.1. Determination of pH

The pH of the formulated gel was measured using a digital pH meter. The electrode was rinsed with distilled water, then immersed in the gel formulation to obtain the pH reading [17].

2.6.2. Spreadability

A modified spreadability apparatus was used to assess the spreadability of the formulations. One gram of gel was placed between two horizontal glass slides (7.5×2.5 cm), and a constant load of 100 g was applied to the upper slide for 1 minute. Spreadability was determined by measuring the time (in seconds) required for the upper slide to move a set distance of 6.5 cm [18].

Spreadability was calculated as,

$$S=M \times L/T$$

Where, S- Spreadability, M- weight attached to upper slide, L- Length of glass slides, T- Separation time of slides.

2.6.3. Viscosity

The viscosity of the gel formulation was measured using a Brookfield viscometer with spindle 95 at 50 rpm, set at either $25 \pm 1.0^\circ\text{C}$ or $32 \pm 1.0^\circ\text{C}$ ¹⁹. The results represent the average values obtained from three different batches.

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2.6.4. Atomic force microscopy

The surface topography and roughness of the gel were analysed using Atomic Force Microscopy (AFM) with a NANOSURF C3000. A small aliquot of the gel was placed on freshly cleaved mica, left to evaporate at room temperature (40°C), and then placed into the AFM's liquid cell. The analysis was conducted in tapping mode with a line scan rate of 1-0.3 Hz and a 30% set point. The spring constant of the point-probe was 54 N/m with a tip radius of <10 nm. Data were acquired from a $20 \times 20 \mu\text{m}^2$ area using Easy Scan 2 software, and roughness parameters such as mean roughness (S_a), root mean square roughness (S_q), and peak height (S_p) were calculated [20].

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2.6.5. Drug content

The colchicine content in the gel formulations was determined using UV-visible spectroscopy. One gram of gel was dissolved in a sufficient amount of phosphate buffer (pH 7.4), filtered, and the volume was adjusted to 50 mL. The absorbance was then measured at 350 nm after appropriate dilution [21].

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26.6. Skin irritation test

Three Wistar rats were housed in metal cages with perforated floors. The room temperature was maintained at $22 \pm 3^\circ\text{C}$, with 30-70% relative humidity. The lighting was controlled to provide 12 hours of artificial light (8 am - 8 pm) each day. Twenty-four hours prior to the test, the fur on the dorsal side of each rat was shaved to expose an area of approximately 5 cm^2 of skin. The final gel formulation was then evenly applied to this exposed 5 cm^2 area of skin. Skin reactions at the application site were subjectively assessed and scored once daily at 1, 24, and 48 hours during the post-test observation period [16,22].

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The fur on the dorsal side of each rat was shaved 24 hours prior to the test to expose approximately 5 cm^2 of skin area.

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2.6.7. Ex-vivo drug diffusion studies

Vertical type of Franz diffusion cells using goat skin was used to perform the *ex-vivo* drug diffusion studies. The previously calibrated volume of the formulation was placed in the donor compartment, and 8 ml phosphate buffer (pH 6.5) was placed in the receptor compartment, the goat skin of thickness of 2.6 mm exposing a surface area of 3.799 cm^2 was used. Subcutaneous fat content was cleaned from the goat skin and placed between the diffusion cells. The *ex-vivo* diffusion studies were stopped after 7 hr since the goat skin loses its viability after a few hours when placed at room temperature. Samples were analysed at 350 nm using a UV Spectrophotometer. The permeation profile was obtained by plotting a graph of cumulative drug permeation versus time²³. The flux ($\mu\text{g}/\text{cm}^2/\text{h}$), and permeability coefficient were calculated from the slope of the linear portion of the cumulative drug permeated ($\mu\text{g}/\text{cm}^2$) versus the time (h) profile.

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The degree of drug release from the considered formulations was compared to the standard capsaicin.

2.6.8 In-vivo studies

Wistar rats of both sexes, weighing 200–300 g, were acclimated to laboratory conditions for 7 days prior to the experiment. The animals were divided into three groups ($n=2$). The fur on each rat's back was carefully shaved, avoiding any skin abrasion. Group I received an oral dose of colchicine (0.5 mg tablet dissolved in distilled water) as the standard. Groups II and III received topical applications of the test formulation SL06 (selected from *ex vivo* diffusion studies). Blood samples were collected from each animal at specific time intervals (0.5, 1, 4, 6, 12, 24 and 48 hours post-administration) under light anaesthesia from the retro-orbital plexus. Samples were drawn into heparinized tubes, and plasma was separated for the estimation of test and standard compound concentrations. Plasma concentrations were determined using an HPLC analytical method.

3. Result

3.1. Identification of critical material attributes

Key material factors were identified for the primary excipients, specifically the concentration of the oil phase (*Simmondsiachinensis* oil) and the gelling agent (Carbopol 940). Table 2 details how laboratory trials were conducted to determine the optimal quantities of these critical material factors.

Table 2 Details of experimental conditions

Independent Variables	Levels		
	Low (-1)	Medium (0)	High (+1)
Conc. of Gelling agent	0.7%	1.7%	2.7%
Conc. of Oil	0.1%	0.5%	0.9%

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3.2. Identification of critical quality attributes

Literature review identified the *in vitro* drug release rate as a critical quality attribute. The *in vitro* release study was performed using a Franz diffusion cell under optimal sink conditions.

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3.3. Preparation and optimization of formulation

Colchicine-loaded gel was prepared using Carbopol 940 as the gelling agent, *Simmondsia chinensis* oil as a natural penetration enhancer forming the oil base, isopropyl alcohol and propylene glycol as solvents, and methyl paraben and propyl paraben as preservatives. The formulation compositions are detailed in Table 3.

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Table 3 Composition and of different formulations suggested by Design Expert software

Std	Run	Factor 1	Factor 2	Response 1	Response 2
		A: Conc. of Gelling agent (%)	B: Conc. of Oil (ml)	Viscosity (cps)	In-vitro Drug Release (%)
1	1	0.7	0.1	10322.04	32.18
7	2	0.7	0.5	15278.28	37.54
4	3	0.7	0.9	32355.01	55.74
8	4	1.7	0.1	39617.71	85.28
6	5	1.7	0.5	43529.61	95.58

3	6	1.7	0.9	59171.17	100
9	7	2.7	0.1	89613.36	51.24
5	8	2.7	0.5	98364.89	53.24
2	9	2.7	0.9	120647.3	59.44

3.4. Implementation of design

The experimental data were analysed using Design Expert® software, allowing for clear interpretation of the relationships between variables and responses through contour and 3D surface plots, as illustrated in Figure 1 and 3.

3.4.1. Counter plot of viscosity (Y1) with variables X1 and X2

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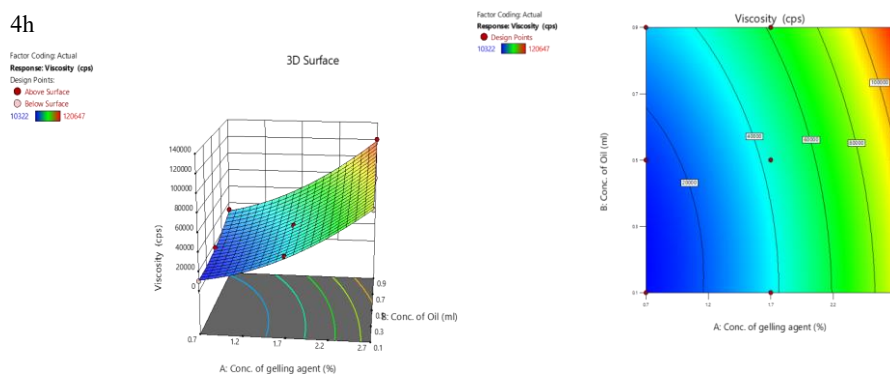


Figure 1 Effect of gelling agent and oil on viscosity (a) 3d surface plot; (b) contour plot

The model for viscosity was significant as p value found less than 0.05. Lack of fit was non-significant for the model. The high value of regression coefficient for the model signifies the goodness of fit as shown in Table 4. The high value to regression coefficient explained the strong relationship between the factors with the response.

Table 4 Results of viscosity ANOVA analysis

Observations	9
Mean of Responses	56544.37
p value	0.0372
R^2	0.9986
Adjusted R^2	0.9963
Predicted R^2	0.9830

The relationship between viscosity, the concentration of the gelling agent, and the concentration of oil is expressed in Equation 1.

$$\text{Viscosity} = 43286.04888889 + 41778.37A + 12103.395B + 2250.2425AB + 13657.316666667A^2 + 6230.1716666667B^2$$

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Where,A: Concentration of Gelling agent;B: Concentration of Oil

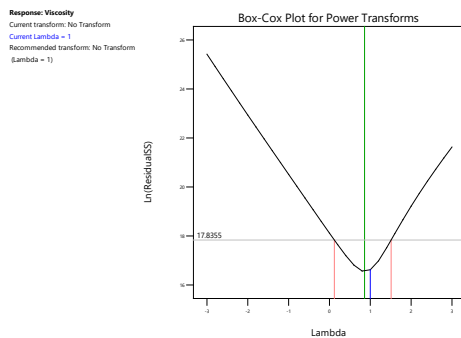


Figure 2Viscosity Box-Cox plot

The Box-Cox plot for the gelling agent's effect on viscosity indicates that no transformation of the data is necessary. This conclusion is based on the current lambda (λ) value being 1, which aligns with the raw data.

3.4.2. Counter plot of drug release (Y2) with variables X1 and X2

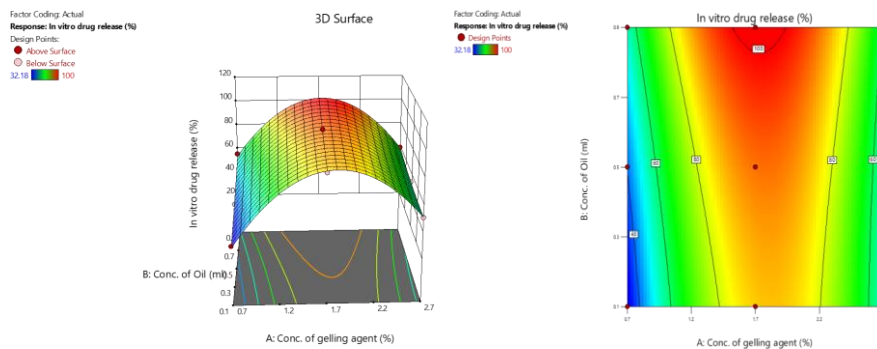


Figure 3Effect of gelling agent and oil on drug release (a) 3d surface plot; (b) contour plot

The model for Drug Release was significant as p value found less than 0.05. Lack of fit was non-significant for the model. The high value of regression coefficient for the model signifies the goodness of fit as shown in Table 5. The high value to regression coefficient explained the strong relationship between the factors with the response.

Table 5Results of viscosity ANOVA analysis

Observations	9
Mean of Responses	63.36
p value	0.0016
R^2	0.9938
Adjusted R^2	0.9836
Predicted R^2	0.9476

The relationship between Drug Release, the concentration of the gelling agent, and the

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concentration of oil is expressed in Equation 1.

$$\text{In vitro drug release} = +92.38 + 6.41A + 7.75B - 3.84AB - 45.39A^2$$

Where, A: Concentration of Gelling agent; B: Concentration of Oil

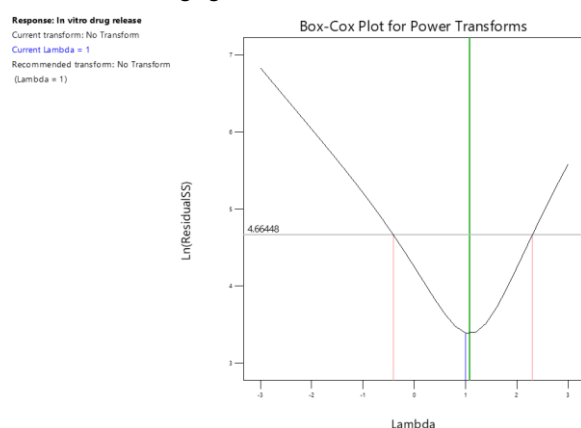


Figure 4 Drug Release Box-Cox Plot

The Box-Cox plot for the gelling agent's effect on *in vitro* drug release indicates that no transformation of the data is necessary. This conclusion is based on the current lambda (λ) value being 1, which aligns with the raw data.

The Design of Experiments (DoE) analysis highlights the critical influence of both the gelling agent and oil concentration on the viscosity and drug release behaviour of the formulation. An increase in gelling agent concentration significantly boosts viscosity by forming a denser polymer network, thereby enhancing the structural integrity of the formulation. However, this higher concentration tends to hinder drug release, as the tighter matrix restricts diffusion. Similarly, oil concentration affects these parameters, with higher levels slightly increasing viscosity due to a thickening effect within the gel matrix. Elevated oil levels, however, can slow drug release by forming a hydrophobic barrier that impedes diffusion. These findings underscore the need to optimize both components to achieve a balance between the desired viscosity and effective drug release.

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3.5. Physiochemical evaluation of gel

3.5.1 Determination of pH

The pH of these formulations ranged from 7.50 ± 0.011 to 7.79 ± 0.017 , indicating suitability for skin application, as the values are close to neutral. This near-neutral pH suggests that the

formulations can be applied to the skin without causing irritation. The pH of all formulations is presented in Table 5.

3.5.2. Spreadability

As displayed in Table 5, the spreadability values of gel formulation were found between 11.01 ± 0.06 to 21.39 ± 0.04 g cm/sec.

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3.5.3. Atomic force microscopy

The analysis revealed a surface roughness of 205.25 nm, with 80% of particles having an average size of 254.79 nm (Figure). This particle size suggests that the particles are well dispersed within the gel matrix, with consistent morphology over a 20-day period. No particle breakage was observed during the 20-day release period. AFM images show that Colchicine, *Simmondsia chinensis* oil, and Carbopol 940 are well-bonded, suggesting a stable formulation that resists rapid degradation.

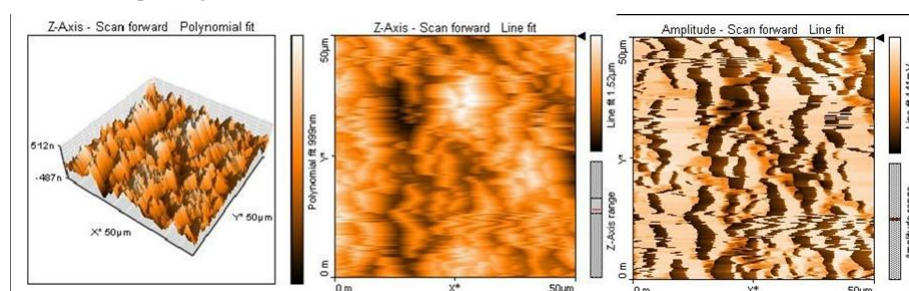


Figure 5 AFM of gel

3.5.4. Drug content

The drug content in the gel ranged from $97 \pm 0.29\%$ to $98.5 \pm 0.28\%$, indicating a high drug-loading efficiency. Details of the drug content are provided in Table 6.

Table 6 Evaluation parameters for gel formulation

Formulation	pH	Spreadability (g cm/sec)	Drug Content (%)
SC1	7.79 ± 0.012	17.40 ± 0.08	98 ± 0.76
SC2	7.95 ± 0.02	18.50 ± 0.02	97 ± 0.5
SC3	7.79 ± 0.017	21.39 ± 0.04	97.5 ± 0.29
SC4	7.50 ± 0.011	18.69 ± 0.07	98 ± 0.58

SC5	7.65±0.012	16.89±0.02	98± 0.59
SC6	7.79±0.017	16.79±0.05	97.5±0.77
SC7	7.65±0.02	16.35±0.03	97± 0.29
SC8	7.65±0.02	11.01±0.06	98.5±0.28
SC9	7.50±0.05	9.65±0.03	98± 0.57

3.5.5. Skin irritation test

The prepared gel was evaluated for its skin irritant effect, where no irritation was observed for all the formulations (Table 7), even after 48 hr of study, indicating that the prepared gel formulation was found to be safe.

Table 7 Primary skin irritation test for gel

Formulation	1 hr	24 hr	48 hr
SC1	0	0	0
SC2	0	0	0
SC3	0	0	0
SC4	0	0	0
SC5	0	0	0
SC6	0	0	0
SC7	0	0	0
SC8	0	0	0
SC9	0	0	0

3.5.6. Ex-vivo drug diffusion studies

The skin permeation of the formulation was evaluated using Franz diffusion cells. The permeability of formulation into goat skin was revealed by sampling from 0 to 7hr. From the Figure which shows the comparatively high drug release when compared with the data shown by capsaicin. When considered the drug release kinetics it was shown that the formulation SC6 follows zero order kinetics which is considered as independent of concentration of drug release with R² value 0.9941.

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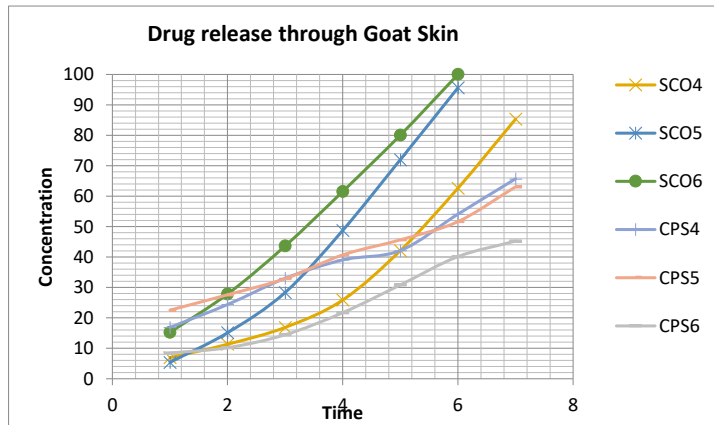


Figure 6 Graph showing drug release of the formulations containing *Simmondsia chinensis* and capsaicin (standard permeation enhancer)

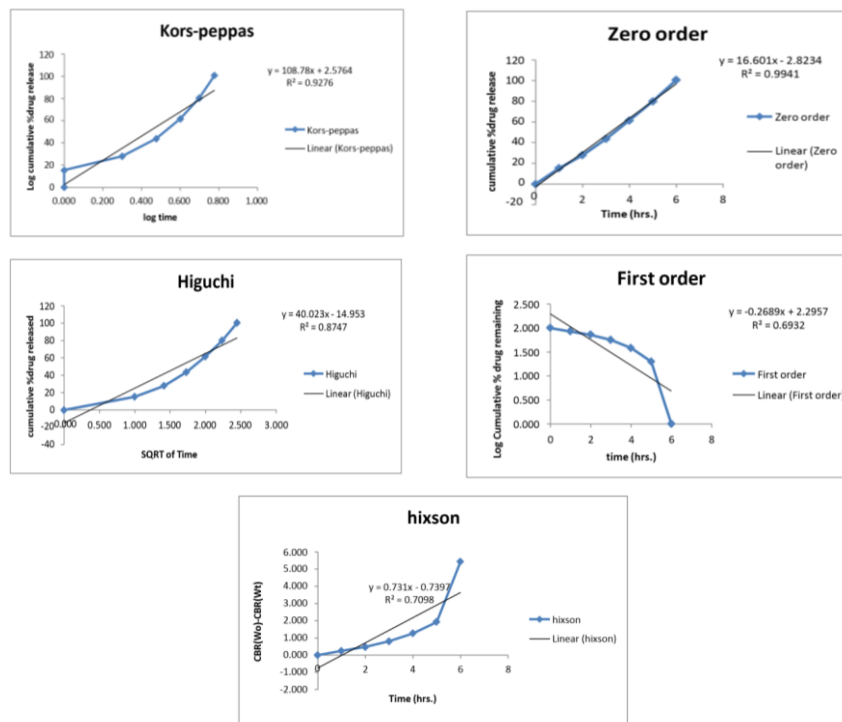


Figure 7 Graph showing mathematical model for the formulation SCO6

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Table 8 Showing regression co-efficient value of SCO6 formulation

Model Name	r^2	SC6	
		Slope	Intercept
Zero order model	0.99	16.60	-2.82
First Order model	0.69	-0.26	2.29
Higuchi model	0.87	40.02	-14.95
Korsmeyer-Peppas model	0.92	108.78	2.57
Hixon-Crowell model	0.70	0.73	-0.73

The mathematical model plays a vital role in the interpretation of mechanism of drug release from a dosage form. It is an important tool to understand the drug release kinetics of a dosage form. The drug release was found to be best fitted by Zero order model $r^2 = 0.9941$. Also, the model Korsmeyer-Peppas power law equation states diffusion, which was evaluated by value, n (release component) which is $r^2=0.9267$ which implies that the drug release from the system follow super case II transport.

3.5.7. In-vivo studies

In this study, plasma concentrations were evaluated in Wistar rats. The visual observations of the animals during the experiment are depicted in Figure 8. A comparative analysis of plasma concentrations between the selected formulation SL06 (identified from *ex vivo* diffusion studies) and standard colchicine is presented in Figure 9 and detailed in Table 9-11.

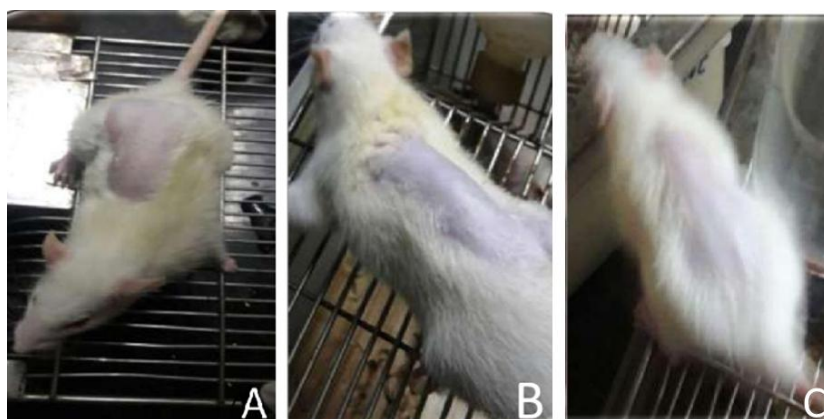


Figure 8 Visual observation of Wister rat during the study: A) application of gel; B) standard drug administration; C) formulation SLO6

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Table 9 Estimation of pharmacokinetic parameters for standard colchicine and the topical formulation SL06 following topical application in Wistar rats

Sr.No.	Parameters	Standard Colchicine	Test sample SL06
1.	Slope	-0.06	-0.016
2.	Intercept	3.69	3.356
3.	Kel (per hour)	0.14	0.037
4.	t1/2(hours)	4.98	18.908
5.	C0(pg./ml)	4910.52	2268.132
6.	Vd(Liters)	101822.27	220445.739
7.	clearance (liters/hours)	14183.45	8079.497
8.	AUC0-t(pg.hr/ml)	3545.99	2019.999
9.	AUC1-t(pg.hr/ml)	44583.17	62347.325
10.	AUC1-∞(pg/ml)	71.79	10460.911
11.	AUC(TOTAL)(pg.hr/ml)	48200.95	74828.235
12.	Cmax(pg/ml)	12000.67	7086.200
13.	Tmax(hour)	1.00	1.000

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Table 10 Pharmacokinetic study standard: Colchicine

Sr. No.	Time(hours)	Concentration(pg/mL)	Log conc.(pg/ml)	AUC
1.	0.5	4000.59	3.602124045	4000.315
2.	1	12000.67	4.079205493	25502.12
3.	4	5000.74	3.699034275	6200.74
4.	6	1200	3.079181246	4800
5.	12	400	2.602059991	3000
6.	24	100	2	1320
7.	48	10	1	-240

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Table 11 Pharmacokinetic Study: SL06

Sr. No.	Time (hours)	Concentration(pg/mL)	Log Conc.	AUC
1.	0.5	178.9	2.25261	1816.275
2.	1	7086.2	3.85041	19388.55
3.	4	5839.5	3.76638	9941.6
4.	6	4102.1	3.61301	15267.9

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5.	12	987.2	2.99441	10793.4
6.	24	811.7	2.90940	14341.2
7.	48	383.4	2.58365	-9201.6

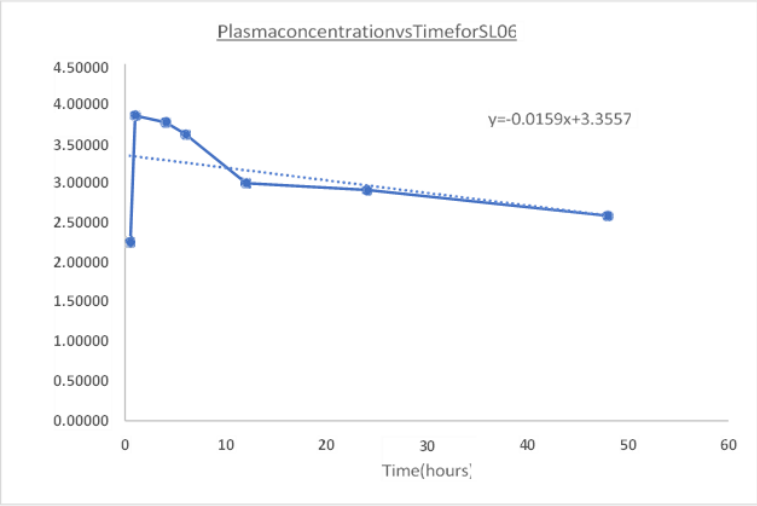


Figure 9In- vivo time vs concentration

4. Discussion

The results of this study highlight the potential of a colchicine-loaded gel with natural permeation enhancers for effective rheumatoid arthritis (RA) management. Using a Quality by Design (QbD) approach, the critical material attributes (concentration of *Simmondsia chinensis* oil and Carbopol 940) and quality attributes (*in vitro* drug release and viscosity) were identified and optimized. The design of experiments (DoE) analysis demonstrated a significant relationship between formulation variables and response parameters, including viscosity and drug release.

The optimized formulation (SL06) achieved a balance between viscosity and drug release. High concentrations of Carbopol 940 increased viscosity, which is essential for the gel’s structural integrity, but excessive levels reduced drug diffusion. Conversely, *Simmondsia chinensis* oil enhanced transdermal delivery by acting as a permeation enhancer while slightly affecting viscosity. The ANOVA analysis confirmed the significance of these relationships, and the mathematical models provided predictive equations for response optimization.

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Physicochemical evaluations affirmed the suitability of the gel for topical application, with near-neutral pH, high spreadability, and excellent drug content. The gel demonstrated stable morphology and sustained drug release, with *ex vivo* diffusion studies revealing superior permeation compared to standard permeation enhancers like capsaicin.

In vivo pharmacokinetic studies further validated the efficacy of the SL06 formulation. It exhibited prolonged half-life ($t_{1/2}$) and enhanced bioavailability compared to standard colchicine, supporting its superior pharmacokinetic profile. The zero-order release kinetics and super case II transport mechanism suggest controlled and sustained drug delivery. Moreover, the absence of skin irritation in primary irritation tests underscores its safety for clinical use.

5. Conclusions

In conclusion, the development of the Colchicine-infused medicated gel with natural permeation enhancers demonstrates promising potential for effective rheumatoid arthritis treatment. The formulation, utilizing *Simmondsia chinensis* oil as a natural penetration enhancer and Carbopol 940 as a gelling agent, ensures optimal drug loading and controlled release. The gel's neutral pH and good skin compatibility further enhance its suitability for topical application. The findings suggest that the gel can effectively deliver Colchicine to the targeted site, providing a non-invasive, localized treatment option with minimal side effects.

References

1. Matteo AD, Bathon JM, Emery P. Rheumatoid arthritis. *Lancet*. 2023;**402**(10416):2019-33.
2. GuoQ, Wang Y, XuD, Nossent J, PavlosNJ, Xu J. Rheumatoid arthritis: Pathological mechanisms and modern pharmacologic therapies. *Bone Res*. 2018;**6**: 15.
3. DingQ, Hu W, WangR, Yang Q, ZhuM, LiM, CaiJ, Rose P, MaoJ, Zhu YZ. Signalling pathways in rheumatoid arthritis: Implications for targeted therapy. *Signal Transduct. Target. Ther*. 2023;**8**: 68.
4. Bullock J, Rizvi SAA, Saleh AM, Ahmed SS, DoDP, Ansari RA, Ahmed J. Rheumatoid Arthritis: A Brief Overview of the Treatment. *Med. Princ. Pract*. 2019;**27**: 501-7.
5. Altman R, Bosch B, Brune K, Patrignani P, Young C. Advances in NSAID Development: Evolution of Diclofenac Products Using Pharmaceutical Technology. *Drugs* 2015; **75**: 859–77.

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6. Ben MR, Bouchmaa N, Ainani H, El Fatimy R, Malka G, Mazini L. Anti-rheumatoid drugs advancements: New insights into the molecular treatment of rheumatoid arthritis. *Biomed. Pharmacother.* 2022; **151**: 113126.
7. Gupta GR, Sharma R. Advancement in the management of rheumatoid arthritis. *RPS Pharm. Pharmacol. Rep.* 2023;**2**: rqad005.
8. Dhule KD, Nandgude TD. Lipid Nano-System Based Topical Drug Delivery for Management of Rheumatoid Arthritis: An Overview. *Adv Pharm Bull.* 2023;**13(4)**:663-77.
9. Phatale V, Vaiphei KK, Jha S, Patil D, Agrawal M, Alexander A. Overcoming skin barriers through advanced transdermal drug delivery approaches. *J Control Release.* 2022; **351**: 361-80.
10. Tawde S, Gawde K, Mahind D, Mhaprolkar C, Sawant K, Surve C. Gingel: Development and Evaluation of Anti-arthritis Gel Containing Ginger (Zingiberofficinale). *IJISRT*2020; **5(7)**: 1148-53.
11. Chlorogiannis DD, Papanagiotou SPP, Bakogiannis NC, Bakoyiannis C, Kokkinidis DG. Inflammation, anti-inflammatory agents, and the role of colchicine in carotid artery stenosis. *Vasa* 2024;**53(1)**:4-12.
12. Bausson J, Keller N, Von Hunolstein JJ. Safety and efficacy of colchicine in crystal-induced arthritis flare in 54 patients with severe chronic kidney disease. *RMD Open* 2024;**10**: e003872.
13. Pradhan R, Krishna KV, WadhwaG, Taliyan R, Khadgawat R, KachhawaG, Singhvi G, DubeySK. QbD-Driven Development and Validation of HPLC Method for Determination of Bisphenol A and Bis-Sulphone in Environmental Samples. *Int. J. Environ. Anal. Chem*2020; **100**: 42–54.
14. Bolla PK, Clark BA, Juluri A, Cheruvu HS, Renukuntla. Journal Evaluation of formulation parameters on permeation of ibuprofen from topical formulations using Strat-M® membrane. *Pharmaceutics* 2020; **12(2)**: 151.
15. ICH. Guidance for Industry: Q8 (R2) Pharmaceutical Development; U.S. Department of Health and Human Services: Washington, DC, USA. 2009; pp. 1–29.
16. Aiyalu R, Govindarjan A, Ramasamy A. Formulation and evaluation of topical herbal gel for the treatment of arthritis in animal model. *BJPS* 2016; **52(3)**: 494-507.
17. Das S, Sharadha M, Venkatesh MP, Sahoo A, Tripathy J, Gowda DV. Formulation And Evaluation of Topical Nanoemulgel of Methotrexate for Rheumatoid Arthritis. *Int J App Pharm*2021; **13(5)**: 351-57.

18. Wadher K, Dabre S, Gaidhane A, Trivedi S, Umekar M. Evaluation of Anti-psoriatic activity of gel containing Pongamiapinnata extract on imiquimod-induced psoriasis. *ClinPhytosci*2021;**7(1)**:1-6.
19. AlnaimA S, Shah H, NairAB, Mewada V, Patel S, Jacob S, Aldhubiab B, MorsyMA ,Almuqbil RM, Shinu P.Qbd-Based Approach to Optimize Niosomal Gel of Levosulpiride for Transdermal Drug Delivery. *Gels* 2023;**9**: 213.
20. Swarna M, JaiganeshR, Rajalakshmanan E. Formulation and evaluation of *Ocimum sanctum* Linn containing carboxymethyl cellulose and sorbitol-based hydrogel. *Bioinformation*2023;**19(5)**: 546-51.
21. Amgaonkar YM, Kochar NI, Chandewar AV, Umekar MJ, Wadher KJ. Boswellic Acid Loaded Nanoemulgel for Rheumatoid Arthritis: Formulation Design and Optimization by QbD, *in vitro*, *ex vivo*, and *in vivo* Evaluation. *Ind. J. Pharm. Edu. Res* 2024; **58(2)**:546-54.
22. DonthiMR, SahaRN, Singhvi G, Dubey SK. Dasatinib-Loaded Topical Nano-Emulgel for Rheumatoid Arthritis: Formulation Design and Optimization by QbD, *in vitro*, *ex vivo*, and *in vivo* Evaluation. *Pharmaceutics* 2023;**15**: 736.
23. Khan AS, Shah KU, Mohaini MA, Alsalman AJ, Hawaj MAA, Alhashem YN. Tacrolimus-loaded solid lipid nanoparticle gel: formulation development and *in vitro* assessment for topical applications. *Gels*. 2022; **18;8(2)**:129.